Experiment 9 Output

Step 1: Check Running Services & Open Ports

sudo ss -tuln

sudo netstat -tulnp

```
root@prathmesh: /home/prathmesh
root@prathmesh:/home/prathmesh# sudo ss -tuln
Netid          State          Recv-Q          Send-Q                  Local Address:Port                      Peer Address:Port          Process
udp            UNCONN         0               0                          127.0.0.54:53                            0.0.0.0:*
udp            UNCONN         0               0                       127.0.0.53%lo:53                            0.0.0.0:*
udp            UNCONN         0               0                             0.0.0.0:48194                         0.0.0.0:*
udp            UNCONN         0               0                             0.0.0.0:5353                          0.0.0.0:*
udp            UNCONN         0               0                                [::]:44247                            [::]:*
udp            UNCONN         0               0                                [::]:5353                             [::]:*
tcp            LISTEN         0               4096                    127.0.0.53%lo:53                            0.0.0.0:*
tcp            LISTEN         0               4096                        127.0.0.1:631                           0.0.0.0:*
tcp            LISTEN         0               4096                       127.0.0.54:53                            0.0.0.0:*
tcp            LISTEN         0               4096                           [::1]:631                            [::]:*
```

```
root@prathmesh:/home/prathmesh# sudo netstat -tulnp
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN      639/systemd-resolve
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN      1159/cupsd
tcp        0      0 127.0.0.54:53           0.0.0.0:*               LISTEN      639/systemd-resolve
tcp6       0      0 ::1:631                 :::*                    LISTEN      1159/cupsd
udp        0      0 127.0.0.54:53           0.0.0.0:*                           639/systemd-resolve
udp        0      0 127.0.0.53:53           0.0.0.0:*                           639/systemd-resolve
udp        0      0 0.0.0.0:48194           0.0.0.0:*                           823/avahi-daemon: r
udp        0      0 0.0.0.0:5353            0.0.0.0:*                           823/avahi-daemon: r
udp6       0      0 :::44247                :::*                                823/avahi-daemon: r
udp6       0      0 :::5353                 :::*                                823/avahi-daemon: r
root@prathmesh:/home/prathmesh#
```

Step 2: Use UFW (Firewall)

sudo ufw enable

sudo ufw default deny incoming

sudo ufw allow 22/tcp

sudo ufw status verbose

```
root@prathmesh:/home/prathmesh# sudo ufw enable
Firewall is active and enabled on system startup
root@prathmesh:/home/prathmesh# sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
root@prathmesh:/home/prathmesh# sudo ufw allow 22/tcp
Rule added
Rule added (v6)
root@prathmesh:/home/prathmesh# sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To                         Action      From
--                         ------      ----
22/tcp                     ALLOW IN    Anywhere
22/tcp (v6)                ALLOW IN    Anywhere (v6)

root@prathmesh:/home/prathmesh#
```

Step 3: Secure User Accounts

- Check users:

cut -d: -f1 /etc/passwd

- Disable unused accounts:

sudo usermod -L username (Here you can delete Alice)

- Set strong passwords:

sudo passwd username (Use the password of Alice)

```
root@prathmesh:/home/prathmesh# cut -d: -f1 /etc/passwd
root
daemon
bin
sys
sync
games
man
lp
mail
news
```

```
nm-openvpn
prathmesh
alice
root@prathmesh:/home/prathmesh# sudo usermod -L alice
root@prathmesh:/home/prathmesh# sudo passwd alice
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
root@prathmesh:/home/prathmesh#
```

Step 4: File and Directory Permissions

- View permissions:

ls -l /etc/shadow

- Modify permissions:

sudo chmod 640 /etc/shadow

sudo chown root:shadow /etc/shadow

Step 5: Rootkit Detection

sudo chkrootkit

sudo rkhunter --check

```
root@prathmesh: /home/prathmesh
Processing triggers for man-db (2.12.0-4build2) ...
root@prathmesh:/home/prathmesh# sudo chkrootkit
ROOTDIR is `/'
Checking `amd'...                                           not found
Checking `basename'...                                      not infected
Checking `biff'...                                          not found
Checking `chfn'...                                          not infected
Checking `chsh'...                                          not infected
Checking `cron'...                                          not infected
Checking `crontab'...                                       not infected
Checking `date'...                                          not infected
Checking `du'...                                            not infected
Checking `dirname'...                                       not infected
Checking `echo'...                                          not infected
Checking `egrep'...                                         not infected
```

```
Searching for sebek LKM (Adore based)...                    not tested
Searching for knark LKM rootkit...                          not found
Searching for for hidden processes with chkproc...          not found
Searching for for hidden directories using chkdirs...       not found
Checking `lkm'...                                           finished
Checking `rexedcs'...                                       not found
Checking `sniffer'...                                       WARNING

WARNING: Output from ifpromisc:
lo: not promisc and no packet sniffer sockets
ens33: PACKET SNIFFER(/usr/sbin/NetworkManager[893])

Checking `w55808'...                                        not found
Checking `wted'...                                          not found
Checking `scalper'...                                       not found
Checking `slapper'...                                       not found
Checking `z2'...                                            not found
Checking `chkutmp'...                                       not found
Checking `OSX_RSPLUG'...                                    not tested
root@prathmesh:/home/prathmesh#
```

```
root@prathmesh: /home/prathmesh
[ Rootkit Hunter version 1.4.6 ]
File updated: searched for 181 files, found 142
root@prathmesh:/home/prathmesh# sudo rkhunter --check
[ Rootkit Hunter version 1.4.6 ]

Checking system commands...

  Performing 'strings' command checks
    Checking 'strings' command                              [ OK ]

  Performing 'shared libraries' checks
    Checking for preloading variables                       [ None found ]
    Checking for preloaded libraries                        [ None found ]
    Checking LD_LIBRARY_PATH variable                       [ Not found ]

  Performing file properties checks
    Checking for prerequisites                              [ OK ]
    /usr/sbin/adduser                                       [ OK ]
    /usr/sbin/chroot                                        [ OK ]
    /usr/sbin/cron                                          [ OK ]
```

```
root@prathmesh: /home/prathmesh
    /usr/bin/mail.mailutils                                 [ OK ]
    /usr/bin/dash                                           [ OK ]
    /usr/bin/x86_64-linux-gnu-size                          [ OK ]
    /usr/bin/x86_64-linux-gnu-strings                       [ OK ]
    /usr/bin/inetutils-telnet                               [ OK ]
    /usr/bin/which.debianutils                              [ OK ]
    /usr/lib/systemd/systemd                                [ OK ]

[Press <ENTER> to continue]


Checking for rootkits...

  Performing check of known rootkit files and directories
    55808 Trojan - Variant A                                [ Not found ]
    ADM Worm                                                [ Not found ]
    AjaKit Rootkit                                          [ Not found ]
    Adore Rootkit                                           [ Not found ]
    aPa Kit                                                 [ Not found ]
    Apache Worm                                             [ Not found ]
    Ambient (ark) Rootkit                                   [ Not found ]
    Balaur Rootkit                                          [ Not found ]
    BeastKit Rootkit                                        [ Not found ]
    beX2 Rootkit                                            [ Not found ]
```

```
root@prathmesh: /home/prathmesh
    zaRwT.KiT Rootkit                                    [ Not found ]
    ZK Rootkit                                           [ Not found ]

[Press <ENTER> to continue]


  Performing additional rootkit checks
    Suckit Rootkit additional checks                     [ OK ]
    Checking for possible rootkit files and directories  [ None found ]
    Checking for possible rootkit strings                [ None found ]

  Performing malware checks
    Checking running processes for suspicious files      [ None found ]
    Checking for login backdoors                         [ None found ]
    Checking for sniffer log files                       [ None found ]
    Checking for suspicious directories                  [ None found ]
    Checking for suspicious (large) shared memory segments [ None found ]
    Checking for Apache backdoor                         [ Not found ]

  Performing Linux specific checks
    Checking loaded kernel modules                       [ OK ]
    Checking kernel module names                         [ OK ]

[Press <ENTER> to continue]


Checking the network...

  Performing checks on the network ports
    Checking for backdoor ports                          [ None found ]

  Performing checks on the network interfaces
    Checking for promiscuous interfaces                  [ None found ]

Checking the local host...
```

```
root@prathmesh: /home/prathmesh
  Performing system configuration file checks
    Checking for an SSH configuration file               [ Not found ]
    Checking for a running system logging daemon         [ Found ]
    Checking for a system logging configuration file     [ Found ]
    Checking if syslog remote logging is allowed         [ Not allowed ]

  Performing filesystem checks
    Checking /dev for suspicious file types              [ None found ]
    Checking for hidden files and directories            [ Warning ]

[Press <ENTER> to continue]



System checks summary
=====================

File properties checks...
    Files checked: 142
    Suspect files: 1

Rootkit checks...
    Rootkits checked : 477
    Possible rootkits: 0

Applications checks...
    All checks skipped

The system checks took: 7 minutes and 50 seconds

All results have been written to the log file: /var/log/rkhunter.log

One or more warnings have been found while checking the system.
Please check the log file (/var/log/rkhunter.log)

root@prathmesh:/home/prathmesh#
```

Step 6: Perform Security Audit with Lynis

sudo lynis audit system

Review suggestions at the end of the audit.

```
root@prathmesh:/home/prathmesh# sudo lynis audit system

[ Lynis 3.0.9 ]

################################################################################
  Lynis comes with ABSOLUTELY NO WARRANTY. This is free software, and you are
  welcome to redistribute it under the terms of the GNU General Public License.
  See the LICENSE file for details about using this software.

  2007-2021, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)
################################################################################


[+] Initializing program
------------------------------------
  - Detecting OS...                                          [ DONE ]
  - Checking profiles...                                     [ DONE ]

  ---------------------------------------------------
  Program version:           3.0.9
  Operating system:          Linux
  Operating system name:     Ubuntu
  Operating system version:  24.04
  Kernel version:            6.11.0
```

```
root@prathmesh: /home/prathmesh

      - libpam-tmpdir                                       [ Not Installed ]
  - File System Checks:
    - DM-Crypt, Cryptsetup & Cryptmount:
  - Software:
    - apt-listbugs                                          [ Not Installed ]
    - apt-listchanges                                       [ Not Installed ]
    - needrestart                                           [ Not Installed ]
    - fail2ban                                              [ Not Installed ]
]

[+] Boot and services
------------------------------------
  - Service Manager                                         [ systemd ]
  - Checking UEFI boot                                      [ DISABLED ]
  - Checking presence GRUB2                                 [ FOUND ]
    - Checking for password protection                      [ NONE ]
  - Check running services (systemctl)                      [ DONE ]
        Result: found 34 running services
  - Check enabled services at boot (systemctl)              [ DONE ]
        Result: found 58 enabled services
  - Check startup files (permissions)                       [ OK ]
  - Running 'systemd-analyze security'
        - ModemManager.service:                             [ MEDIUM ]
        - NetworkManager.service:                           [ EXPOSED ]
        - accounts-daemon.service:                          [ MEDIUM ]
        - alsa-state.service:                               [ UNSAFE ]
        - anacron.service:                                  [ UNSAFE ]
        - avahi-daemon.service:                             [ UNSAFE ]
        - colord.service:                                   [ PROTECTED ]
        - cron.service:                                     [ UNSAFE ]
```

```
root@prathmesh: /home/prathmesh

  Hardening index : 66 [#############       ]
  Tests performed : 254
  Plugins enabled : 1

  Components:
  - Firewall               [V]
  - Malware scanner        [V]

  Scan mode:
  Normal [V]  Forensics [ ]  Integration [ ]  Pentest [ ]

  Lynis modules:
  - Compliance status      [?]
  - Security audit         [V]
  - Vulnerability scan     [V]

  Files:
  - Test and debug information      : /var/log/lynis.log
  - Report data                     : /var/log/lynis-report.dat

================================================================================

  Lynis 3.0.9

  Auditing, system hardening, and compliance for UNIX-based systems
  (Linux, macOS, BSD, and others)

  2007-2021, CISOfy - https://cisofy.com/lynis/
  Enterprise support available (compliance, plugins, interface and tools)

================================================================================

  [TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)

root@prathmesh:/home/prathmesh#
```

Step 7: Configure System Logging and Auditing

sudo auditctl -e 1

sudo systemctl enable auditd

To monitor file:

sudo auditctl -w /etc/passwd -p war -k passwd_changes

To view logs:

sudo ausearch -k passwd_changes

```
root@prathmesh:/home/prathmesh# sudo auditctl -e 1
enabled 1
failure 1
pid 207632
rate_limit 0
backlog_limit 8192
lost 0
backlog 0
backlog_wait_time 60000
backlog_wait_time_actual 0
root@prathmesh:/home/prathmesh# sudo systemctl enable auditd
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
root@prathmesh:/home/prathmesh# sudo auditctl -w /etc/passwd -p war -k passwd_changes
root@prathmesh:/home/prathmesh# sudo ausearch -k passwd_changes
----
time->Sat Mar 29 12:23:26 2025
type=PROCTITLE msg=audit(1743231206.621:302): proctitle=617564697463746C002D77002F6574632F706173737764002D7000776172002D6B007061737377645F6368616E676573
type=SYSCALL msg=audit(1743231206.621:302): arch=c000003e syscall=44 success=yes exit=1084 a0=4 a1=7ffe8caaf770 a2=43c a3=0 items=0 ppid=208189 pid=208190 au
id=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts2 ses=3 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(1743231206.621:302): auid=1000 ses=3 subj=unconfined op=add_rule key="passwd_changes" list=4 res=1
----
time->Sat Mar 29 12:23:28 2025
```

```
time->Sat Mar 29 12:23:38 2025
type=PROCTITLE msg=audit(1743231218.131:317): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1743231218.131:317): item=0 name="/etc/passwd" inode=264988 dev=08:02 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1743231218.131:317): cwd="/home/prathmesh"
type=SYSCALL msg=audit(1743231218.131:317): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7927fc9cf320 a2=80000 a3=0 items=1 ppid=3068 pid=208
192 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=3 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
----
time->Sat Mar 29 12:23:38 2025
type=PROCTITLE msg=audit(1743231218.132:318): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1743231218.132:318): item=0 name="/etc/passwd" inode=264988 dev=08:02 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1743231218.132:318): cwd="/home/prathmesh"
type=SYSCALL msg=audit(1743231218.132:318): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7927fc9cf320 a2=80000 a3=0 items=1 ppid=3068 pid=208
192 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=3 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
----
time->Sat Mar 29 12:23:38 2025
type=PROCTITLE msg=audit(1743231218.132:319): proctitle=7375646F006175736561726368002D6B007061737377645F6368616E676573
type=PATH msg=audit(1743231218.132:319): item=0 name="/etc/passwd" inode=264988 dev=08:02 mode=0100644 ouid=0 ogid=0 rdev=00:00 nametype=NORMAL cap_fp=0 cap_
fi=0 cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(1743231218.132:319): cwd="/home/prathmesh"
type=SYSCALL msg=audit(1743231218.132:319): arch=c000003e syscall=257 success=yes exit=13 a0=ffffff9c a1=7927fc9cf320 a2=80000 a3=0 items=1 ppid=3068 pid=208
192 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=3 comm="sudo" exe="/usr/bin/sudo" subj=unconfined key="passwd_changes"
root@prathmesh:/home/prathmesh#
```

Step 8: Install and Scan with ClamAV Antivirus

sudo freshclam

sudo clamscan -r /home

```
root@prathmesh: /home/prathmesh
root@prathmesh:/home/prathmesh# ps aux | grep freshclam
clamav    208818  4.2  0.5 162412 20920 ?        Ss   12:24   0:05 /usr/bin/freshclam -d --foreground=true
root      209602  0.0  0.0   9276  2140 pts/1    S+   12:27   0:00 grep --color=auto freshclam
root@prathmesh:/home/prathmesh# sudo systemctl stop clamav-freshclam
root@prathmesh:/home/prathmesh# sudo freshclam
ClamAV update process started at Sat Mar 29 12:28:09 2025
Sat Mar 29 12:28:09 2025 -> daily.cvd database is up-to-date (version: 27591, sigs: 2074276, f-level: 90, builder: raynman)
Sat Mar 29 12:28:09 2025 -> main.cvd database is up-to-date (version: 62, sigs: 6647427, f-level: 90, builder: sigmgr)
Sat Mar 29 12:28:09 2025 -> bytecode.cvd database is up-to-date (version: 336, sigs: 83, f-level: 90, builder: nrandolp)
root@prathmesh:/home/prathmesh# sudo systemctl start clamav-freshclam
root@prathmesh:/home/prathmesh# sudo clamscan -r /home
Loading:    17s, ETA:   0s [=========================>]    8.71M/8.71M sigs
Compiling:   2s, ETA:   0s [=========================>]       41/41 tasks

/home/alice/.bashrc: OK
/home/alice/.profile: OK
/home/alice/.bash_logout: OK
/home/prathmesh/.bashrc: OK
/home/prathmesh/.profile: OK
/home/prathmesh/.bash_logout: OK
/home/prathmesh/.sudo_as_admin_successful: Empty file
/home/prathmesh/.bash_history: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-05-37.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 11-55-08.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-21-54.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-14-48.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-15-02.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-23-45.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 11-59-09.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-03-50.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 11-54-54.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 11-57-30.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-03-11.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-17-47.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 11-58-08.png: OK
/home/prathmesh/Pictures/Screenshots/Screenshot from 2025-03-29 12-22-56.png: OK
```

```
root@prathmesh: /home/prathmesh
/home/prathmesh/snap/firmware-updater/127/.config/user-dirs.dirs.md5sum: OK
/home/prathmesh/snap/firmware-updater/127/.config/user-dirs.locale: OK
/home/prathmesh/snap/firmware-updater/127/.config/user-dirs.locale.md5sum: OK
/home/prathmesh/snap/firmware-updater/127/.last_revision: OK
/home/prathmesh/snap/firmware-updater/127/.local/share/themes: Symbolic link
/home/prathmesh/.local/state/wireplumber/restore-stream: OK
/home/prathmesh/.local/share/nautilus/tags/meta.db: OK
/home/prathmesh/.local/share/nautilus/tags/meta.db-shm: OK
/home/prathmesh/.local/share/nautilus/tags/ontologies.gvdb: OK
/home/prathmesh/.local/share/nautilus/tags/meta.db-wal: Empty file
/home/prathmesh/.local/share/nautilus/tracker2-migration-complete: Empty file
/home/prathmesh/.local/share/gnome-settings-daemon/input-sources-converted: Empty file
/home/prathmesh/.local/share/keyrings/user.keystore: OK
/home/prathmesh/.local/share/keyrings/login.keyring: OK
/home/prathmesh/.local/share/session_migration-ubuntu: OK
/home/prathmesh/.local/share/evolution/tasks/system/tasks.ics: OK
/home/prathmesh/.local/share/evolution/addressbook/system/contacts.db: OK
/home/prathmesh/.local/share/gnome-shell/application_state: OK
/home/prathmesh/.local/share/gnome-shell/update-check-46: Empty file
/home/prathmesh/.local/share/gvfs-metadata/root-44d05e88.log: OK
/home/prathmesh/.local/share/gvfs-metadata/home-b667bf27.log: OK
/home/prathmesh/.local/share/gvfs-metadata/root: OK
/home/prathmesh/.local/share/gvfs-metadata/home: OK

----------- SCAN SUMMARY -----------
Known viruses: 8706051
Engine version: 1.0.8
Scanned directories: 130
Scanned files: 138
Infected files: 0
Data scanned: 9.62 MB
Data read: 24.32 MB (ratio 0.40:1)
Time: 26.261 sec (0 m 26 s)
Start Date: 2025:03:29 12:28:50
End Date:   2025:03:29 12:29:16
root@prathmesh:/home/prathmesh#
```